MADAN MOHAN MALAVIYA UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 7

COURSE: MACHINE LEARNING (BCS-701)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: BCS-701

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Differentiate between supervised, unsupervised, and reinforcement learning.

Q1. (b) Explain bias-variance tradeoff with graphical illustration.

Q1. (c) What is cost function and gradient descent optimization in linear regression?

Q1. (d) Define precision, recall, F1-score, and ROC-AUC curve in classification.

Q1. (e) What is difference between L1 regularization (Lasso) and L2 regularization (Ridge)?

Q1. (f) What is maximum margin hyperplane in Support Vector Machines (SVM)?

Q1. (g) Explain Kernel Trick in SVM and name two popular kernel functions.

Q1. (h) What is curse of dimensionality? How does PCA alleviate it?

Q1. (i) Differentiate between Bagging and Boosting ensemble methods.

Q1. (j) What is backpropagation in Multi-Layer Perceptron (MLP) neural networks?

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Derive gradient descent update rule for parameters theta_0 and theta_1 in Univariate Linear Regression.

Q2. (b) Construct Decision Tree using ID3 algorithm with Information Gain and Entropy for given training dataset.

Q2. (c) Apply K-Means clustering algorithm on 1D points: {2, 4, 10, 12, 3, 20, 30, 11, 25} with k = 2 for 3 iterations.

Q2. (d) Derive backpropagation gradient equations for single hidden-layer neural network with sigmoid activation.

Q2. (e) Explain Principal Component Analysis (PCA) step-by-step: Covariance matrix, eigenvalues, eigenvectors, projection.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain Logistic Regression, sigmoid activation function, and derive cross-entropy loss function. [10 Marks]

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(b) Explain Naive Bayes classifier, Bayes theorem, and Laplace smoothing for text classification. [10 Marks]

Q4. (a) Explain Soft-Margin SVM with slack variables and dual Lagrangian formulation. [10 Marks]

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(b) Explain Random Forest algorithm: Bootstrap aggregation, random feature subspace, out-of-bag error. [10 Marks]

Q5. (a) Explain AdaBoost (Adaptive Boosting) algorithm step-by-step with sample weight updates. [10 Marks]

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(b) Explain Gradient Boosting and XGBoost (Extreme Gradient Boosting) with tree pruning. [10 Marks]

Q6. (a) Explain Hierarchical Clustering: Agglomerative vs Divisive, and Single, Complete, Average linkage. [10 Marks]

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(b) Explain DBSCAN (Density-Based Spatial Clustering) algorithm: Core points, border points, and noise. [10 Marks]

Q7. (a) Explain K-Fold Cross Validation, Stratified K-Fold, and Hyperparameter Tuning with GridSearch. [10 Marks]

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(b) Explain vanishing gradient problem and how ReLU, Leaky ReLU, and batch normalization resolve it. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****